

# Pico WiFi DMX User Manual

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This manual explains how to use the browser-based DMX controller with the Pico firmware. It is written for daily operation: creating fixtures, controlling lights, saving scenes, building chasers, creating motion effects, and using GPIO buttons or ADC inputs.

The web interface is normally opened from XAMPP:

```
http://localhost/dmx/
```

The Pico itself is controlled over the network with its base URL, for example:

```
http://192.168.0.24/
```

Enter the Pico base URL once in any page. The browser stores it and shares it with the other pages.

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## Main Pages

| Page               | Purpose                                                                                |
|--------------------|----------------------------------------------------------------------------------------|
| Fixture Controller | Fixture profiles, patching, live control, groups, scenes, default and blackout values  |
| Chaser             | Step-based chases with fade, loop modes, direction, pause/resume, and Pico slot upload |
| Motion FX          | Pan/tilt effects such as circle, figure-8, pan swing, and tilt swing                   |
| Fan Out            | Spread one control value across a group of fixtures                                    |
| GPIO Control       | Map Pico GPIO buttons and ADC inputs to lighting actions                               |
| Benchmark          | Test Pico HTTP/DMX update performance                                                  |

# 1. Fixture Controller

## DMX Fixture Controller

Setup loaded from server

Pico base URL

☒ Live send

Save setup

Load setup

Motion FX →

Chaser →

Fan Out →

GPIO →

Export CSV

Scenes

Fixture Profiles

Add Control To Selected Profile

Patch Fixtures

### Saved Groups

MAC 4 fixtures

Load

### Control Surface

**MAC 1** Martin MAC XENON · 14 · start 1

Default Blackout Select ▼

Dimmer 0 CH 1

Pan/Tilt 16-bit Pan 2/3 · Tilt 4/5

**MAC 2** Martin MAC XENON · 14 · start 17

Default Blackout Select ▼

Dimmer 0 CH 17

Pan/Tilt 16-bit Pan 18/19 · Tilt 20/21

**MAC 3** Martin MAC XENON · 14 · start 33

Default Blackout Select ▼

Dimmer 0 CH 33

Pan/Tilt 16-bit Pan 34/35 · Tilt 36/37

The Fixture Controller is the central page. Use it first when setting up a new show or fixture set.

## Set the Pico Base URL

1. Open `http://localhost/dmx/`.
2. Enter the Pico base URL, for example `http://192.168.0.24/`.
3. Enable **Live send** when you want control changes to be sent immediately to the Pico.
4. Use **Save setup** to store the fixture setup on the XAMPP server.

# Create a Fixture Profile

The screenshot displays the 'DMX Fixture Controller' web interface. At the top, there's a header with the title 'DMX Fixture Controller' and a subtitle 'Setup loaded from server'. To the right, a 'Pico base URL' field is set to 'http://192.168.0.24/'. Below this are buttons for 'Motion', 'Chaser', 'Fan', 'GPIO', and 'Manual'. A main toolbar contains 'Live send', 'Save setup', 'Load setup', download/upload icons, and a 'Scenes' button with a plus icon. The 'Fixture Profiles' panel on the left lists two profiles: 'Martin MAC XENON' (Mode 14, 13 channels, 6 controls) and 'RGB Spot' (Mode 16ch, 16 channels, 3 controls). Each profile has a list of controls with 'Edit' and 'Remove' buttons. The 'Add / Edit Control' panel on the right is active, showing fields for 'Type, Label & Channels' (e.g., 'Pan/Tilt 16-bit XY', 'Pan/Tilt 16-bit', '1', '3') and 'Default & Blackout' values (e.g., '32768') with corresponding grid visualizations.

A fixture profile describes the DMX personality of one fixture type.

1. Open **Fixture Profiles**.
2. Enter a profile name, mode name, and channel count.
3. Click **Add profile**.
4. Select the profile.
5. Use **Add / Edit Control** to add controls such as dimmer, pan/tilt, RGB, RGBW, RGBWA, wheel, or 16-bit slider.

Each control stores its own DMX channel mapping. For example, a moving head can have:

- Dimmer on channel 1
- Pan/Tilt 16-bit on channels 2/3 and 4/5
- RGBWA color on channels 6-10
- Gobo wheel on channel 12

To change an existing control, click **Edit** in the Fixture Profiles list. The **Add / Edit Control** card opens automatically and loads the selected control. After editing, click **Save control**. The **Fixture Profiles** collapse button also controls the Add / Edit Control card, so both profile editing areas can be hidden together.

## Default and Blackout Values

Each control can store a **Default** value and a **Blackout** value.

Default values are useful for a normal starting look, for example:

- Dimmer open
- Pan/Tilt centered

- RGB white
- Gobo open

Blackout values are useful for quickly shutting down output, for example:

- Dimmer 0
- RGB black
- White and amber 0
- Pan/Tilt at a safe position if needed

For RGB, RGBW, and RGBWA controls, use the color picker for RGB. White and amber channels are edited separately when the fixture has them.

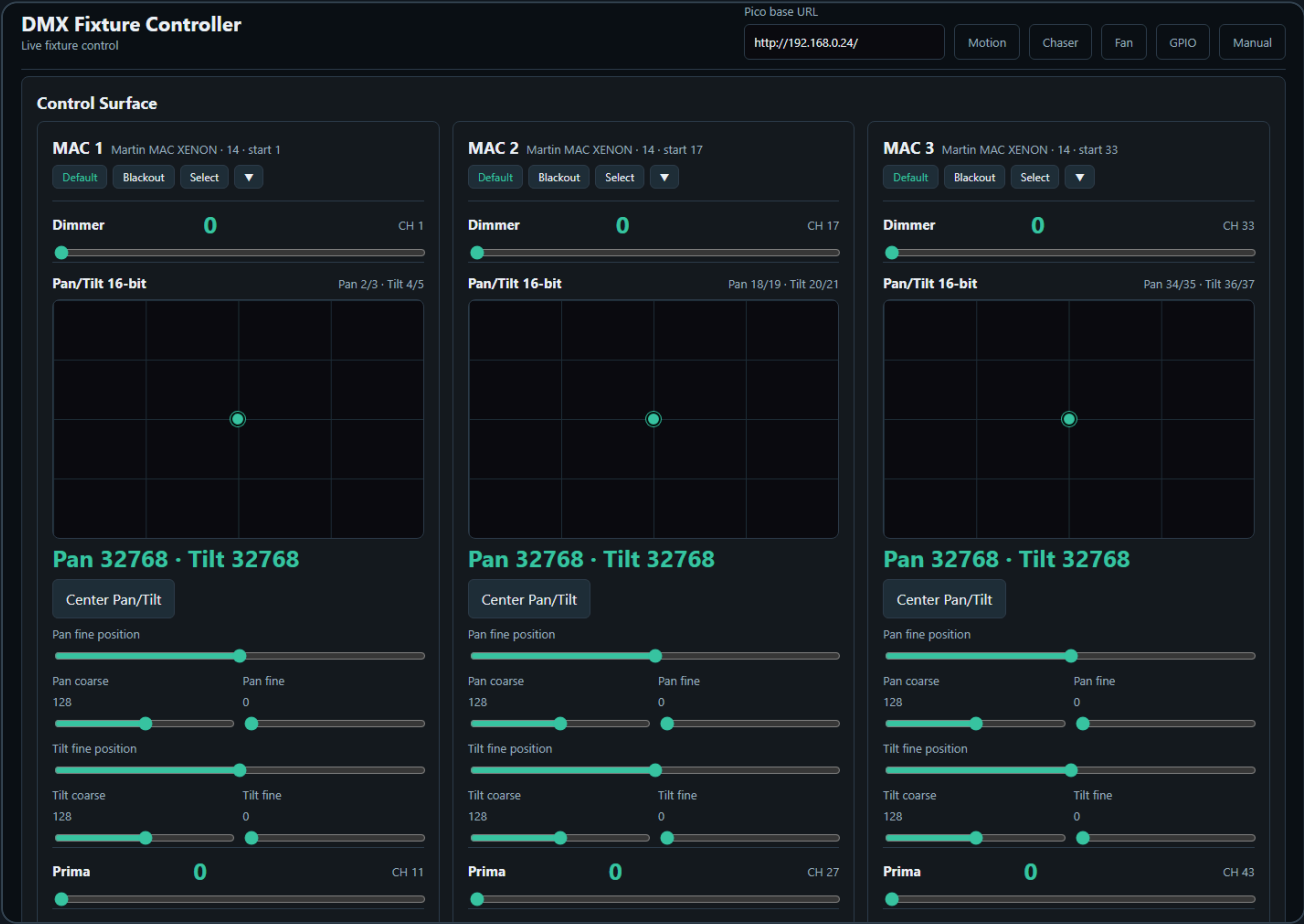
## Patch Fixtures

Patch fixtures after the profile is ready.

1. Open **Patch Fixtures**.
2. Enter a fixture name.
3. Select the fixture profile.
4. Enter the DMX start address.
5. Click **Patch fixture**.

Use **Next free** to find the next available address after already patched fixtures.

## Live Fixture Control



The **Control Surface** shows one card per patched fixture.

Each fixture card contains the controls from its profile:

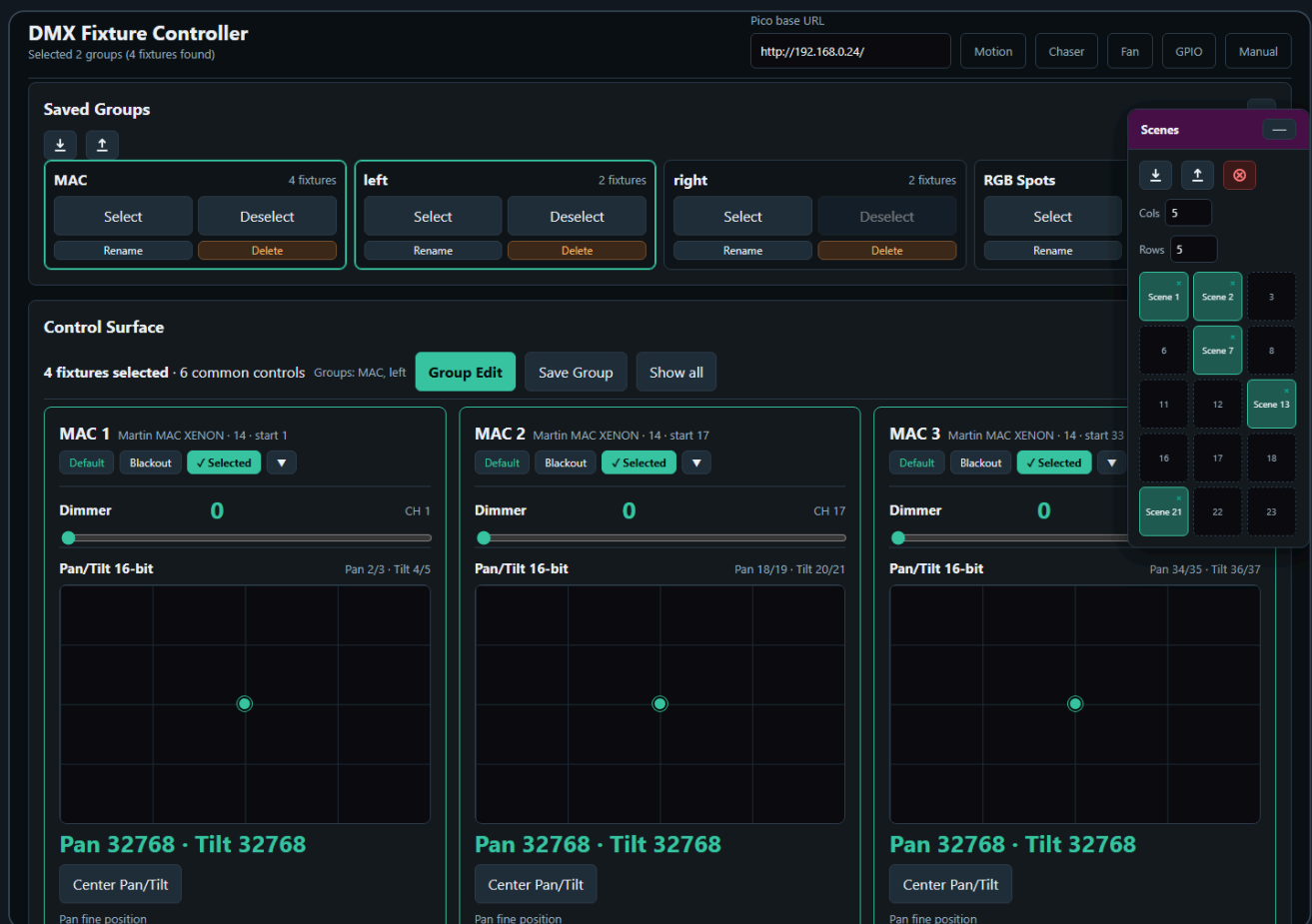
- Sliders for dimmer and simple channels
- XY pads for pan/tilt
- Color pickers and swatches for color controls
- Wheel buttons for indexed wheel values
- Coarse/fine sliders for 16-bit channels

Use **Default** or **Blackout** on a fixture card to recall the stored values for that fixture only.

Use **Select** to include a fixture in group editing.

When you manually select fixture cards, the control surface stays visible so you can keep building or adjusting the selection. Saved Groups can also be selected from the Saved Groups card to filter the control surface.

## 2. Scenes



The Scene Toolbox is available on the Fixture Controller and Fan Out pages.

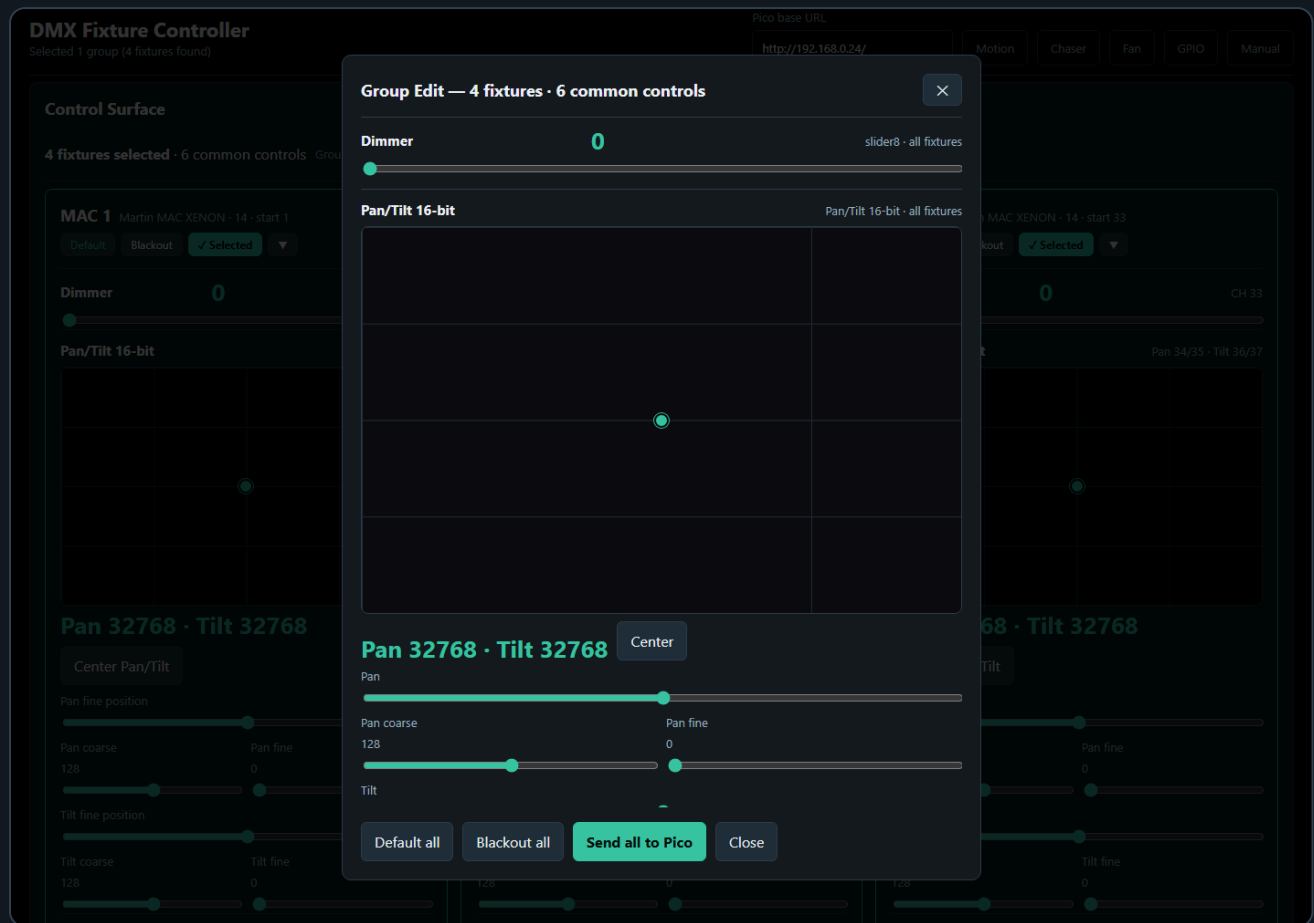
Use scenes to store complete looks.

1. Set your desired fixture values.
2. Click an empty scene slot.
3. Enter a scene name.
4. Click a filled slot to recall it.
5. Use the small **x** on a filled slot to delete it.

Scene recall sends the stored DMX values to the Pico in one batch and updates the live-value snapshot used by the Chaser page.

The red **Clear all DMX channels** button clears controller values and calls `/dmx/clear` on the Pico. This clears both live DMX output and the motion base buffer.

### 3. Groups

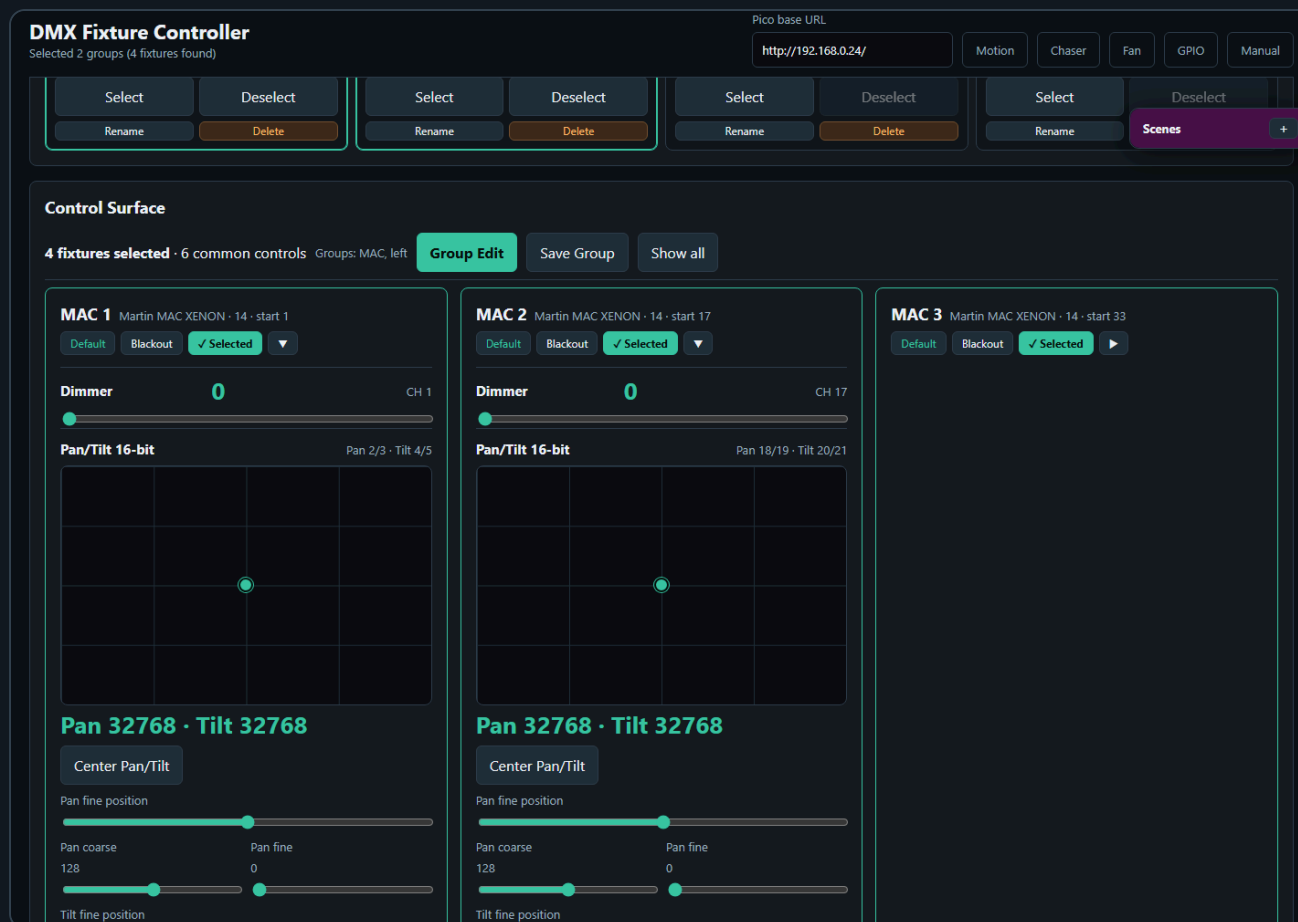


Groups let you control multiple fixtures at once.

#### Create a Group

1. Select two or more fixture cards.
2. Click **Save Group**.
3. Enter a group name.
4. The group appears in **Saved Groups**.

# Saved Group Selection



The Saved Groups card shows groups in a compact matrix. Each group has four actions:

- **Select** adds the group to the active group filter.
- **Deselect** removes only that group from the active filter.
- **Rename** changes the saved group name.
- **Delete** removes the saved group.

More than one saved group can be selected at the same time. The control surface shows the union of all fixtures from the selected groups. This makes it possible to work with several fixture groups together without editing the group definitions.

The group bar above the control surface shows how many fixtures are selected and which saved groups are active. Use **Show all** to clear the saved-group filter and return to the full fixture list.

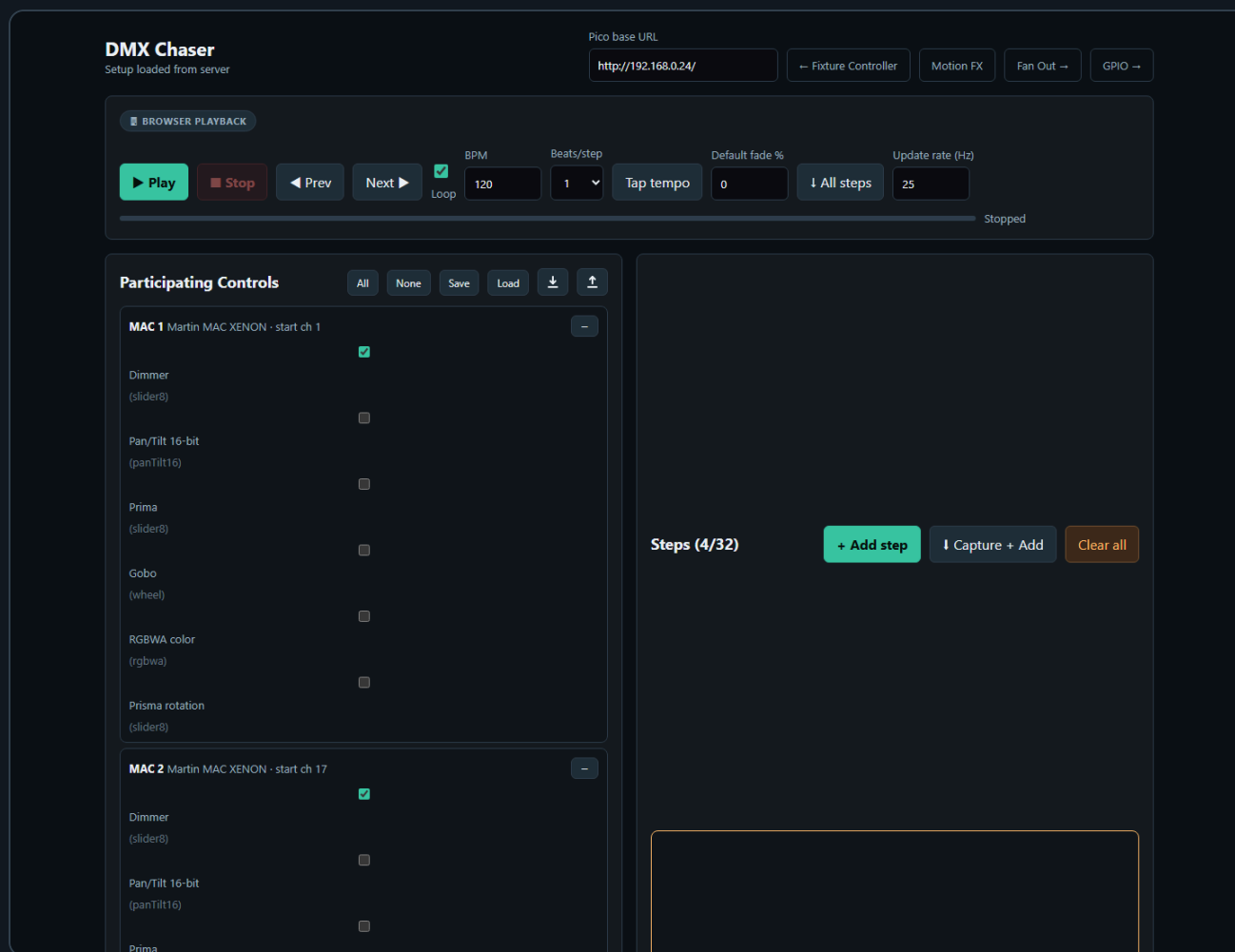
## Edit a Group

1. Load a saved group or select multiple fixtures.
2. Click **Group Edit**.
3. Adjust common controls in the modal.
4. Click **Send all to Pico** to send the group values.

The Group Edit modal only shows controls that exist on all selected fixtures. This prevents accidentally sending values to fixtures with incompatible control layouts.

Use **Default all** or **Blackout all** to recall the stored default or blackout values for every fixture in the selected group.

## 4. Chaser



The Chaser page creates step-based sequences.

### Basic Workflow

1. Open **Chaser**.
2. Select participating fixture controls.
3. Create steps manually or capture values from the Fixture Controller.
4. Set step duration and fade.
5. Save the preset.
6. Upload the preset to a Pico slot.
7. Play the slot from the Pico.

### Participating Controls

Participating controls define which fixture controls belong to the chase. This keeps the chaser from editing unrelated channels.

For example, a dimmer chase might include only dimmer controls. A color chase might include only RGB or RGBWA controls.

### Capture From Fixture Controller

Use **Capture from FC** or **Capture + Add** to read the current Fixture Controller live values and use them as chase step values.



This is useful when you want to build a chase visually:

1. Create a look on the Fixture Controller.
2. Capture it into the Chaser.
3. Change the look.
4. Capture the next step.

## Pico Slot Playback

Chasers can be uploaded to 32 Pico slots. Each slot can run on the Pico without the browser staying open.

Supported playback options:

- Single run
- Loop
- Loop N times
- Forward or reverse direction
- Speed multiplier
- Pause and resume

Each chaser slot supports up to 32 steps.

## 5. Motion FX

DMX Motion Effects

Loading setup...

Pico base URL

http://192.168.0.24/

Fixture Controller

Chaser →

Fan Out →

GPIO →

Effect Parameters

Effect: Circle

BPM: 30

Pan amplitude: 20%

Tilt amplitude: 15%

Phase spread: 0°

Motion path preview (normalized)

MAC 1

MAC 2

MAC 3

MAC 4

Center positions are loaded from the Fixture Controller set. Adjust them per-fixture below without affecting the static setup.

Phase spread offsets fixtures evenly around the cycle — e. 360° with 3 fixtures gives 0°, 120°, 240°, creating a wave effect. Individual phase offsets stack on top.

Scenes

Click a scene to use as motion center

Scene 1

Scene 2

3

Scene 4

Scene 5

6

7

8

Scene 9

Scene 10

Scene 11

12

Scene 13

Scene 14

15

16

17

18

Scene 19

20

Scene 21

22

23

Scene 24

25

Fixtures with Pan/Tilt

MAC 1

16-bit · start ch 1

Live position

Pan 32768 · Tilt 32768

MAC 2

16-bit · start ch 17

Live position

Pan 32768 · Tilt 32768

MAC 3

16-bit · start ch 33

Live position

Pan 32768 · Tilt 32768

MAC 4

16-bit · start ch 49

Live position

Pan 32768 · Tilt 32768

Reload from Fixture Controller

Save Preset

Load Preset

↓

↑

PICO MOTION

Slot 0

BPM 30

Upload to Slot

Start Slot

Set BPM

Stop Slot

Stop All

Restore Saved Slots to Pico

Pico slots: ↓ ↑

Motion FX creates continuous movement for moving lights.

Supported effects include:

- Circle
- Figure-8
- Pan swing
- Tilt swing

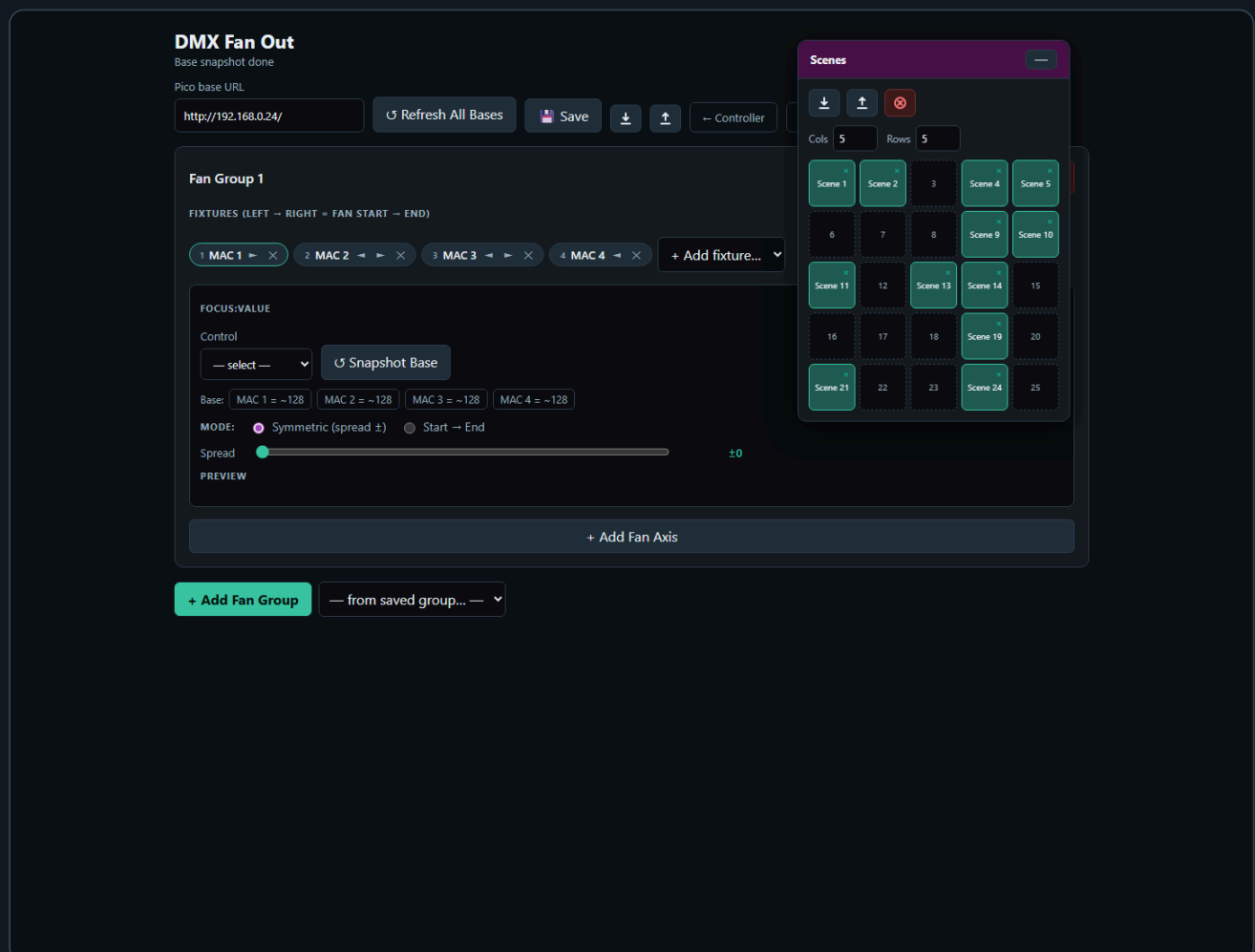
Motion FX is relative to the current scene position. This means the effect moves around the position that was last written into the Pico base buffer.

## Recommended Workflow

1. Use the Fixture Controller to position the fixtures.
2. Save or recall a scene.
3. Open Motion FX.
4. Select the fixtures for the effect.
5. Set BPM, pan amplitude, tilt amplitude, and spread.
6. Upload the effect to a Pico slot.
7. Start the slot.

The Motion FX page also has a read-only scene toolbox. Clicking a scene sends the position to the Pico and updates the effect center.

## 6. Fan Out



Fan Out spreads one selected control across an ordered fixture group.

Typical uses:

- Pan fan
- Tilt fan
- Zoom spread
- Dimmer gradient
- Any other offset across fixtures

## Basic Workflow

1. Open **Fan Out**.
2. Create or select a fan group.
3. Add fixtures in the desired order.
4. Select the control to fan, for example Pan.
5. Snapshot or refresh the base values.
6. Adjust the spread.

Fan Out adds offsets on top of the current base values. This keeps the fan relative to the current fixture positions.

The Fan Out page also includes the scene toolbox, so fan looks can be saved and recalled as scenes.

## 7. GPIO Control

GPIO Control

Ready

Pico base URL

http://192.168.1.50

Controller

Chaser

Motion

Fan Out

GPIO Mappings

not connected

GPIO polling enabled

Add mapping

Add ADC speed/BPM

Push to Pico

Read from Pico

Save local

Export JSON

Import JSON

Clear local

Mapping 1

Remove

GPIO pin

Pull

Trigger

Action

GPIO14

Pull-up

Falling

dmx clear

Slot

Debounce ms

0

30

Mapping 2

Remove

GPIO pin

Pull

Trigger

Action

GPIO15

Pull-up

Falling

chaser toggle

Slot

Debounce ms

0

30

Status

Pico GPIO status

No status yet.

Generated config body

ENABLE 1  
MAP 14 pullup falling dmx\_clear 0 30  
MAP 15 pullup falling chaser\_toggle 0 30

GPIO Control maps physical Pico inputs to lighting actions.

Digital GPIO pins can trigger:

- DMX clear
- Output-only clear
- Stop all
- Chaser play, stop, toggle, pause, resume, pause toggle
- Chaser tap tempo
- Motion start, stop, toggle
- Motion tap tempo

ADC pins can control:

- Chaser speed multiplier
- Motion FX BPM

Only GPIO26, GPIO27, and GPIO28 support ADC input on the Pico 2 W.

## Add a Digital Button Mapping

1. Open **GPIO Control**.
2. Click **Add mapping**.
3. Select a free GPIO pin.
4. Choose pull mode and trigger edge.
5. Choose the action.
6. Set the slot number if the action needs one.
7. Upload the config to the Pico.

The page disables reserved pins and pins already used by other mappings.

## Add an ADC Mapping

1. Add an ADC mapping.
2. Select GPIO26, GPIO27, or GPIO28.
3. Choose `chaser_speed` or `motion_bpm`.
4. Set the target slot.
5. Set the min and max value.
6. Upload the config.

ADC values are filtered on the firmware side with a short mean filter to reduce ripple.

## Tap Tempo

Tap actions use the time between button presses.

The beat divider can be set to:

- 1 beat
- 1/2 beat
- 1/4 beat

- 1/8 beat
- 1/16 beat

The firmware ignores very long gaps between taps so stopping for a while does not create an extremely slow tempo when tapping resumes.

## 8. Benchmark

**DMX Frame Rate Test**  
Configure and press Start.

Pico base URL  
http://192.168.1.50

Preset: Quick check ▾ Start channel: 1 Value: 128 Channels / request: 1 Request count: 200 Duration s: 0

Start Stop Export CSV

single GET /dmx/set/{ch}/{val}

|                       |                                         |                             |              |                 |                        |                   |
|-----------------------|-----------------------------------------|-----------------------------|--------------|-----------------|------------------------|-------------------|
| Throughput<br>req / s | Channels / s<br>effective DMX updates/s | Avg latency<br>ms / request | Median<br>ms | p95 / p99<br>ms | Jitter<br>p95 - median | Min latency<br>ms |
| Max latency<br>ms     | Completed                               | Errors<br>0 failed requests |              |                 |                        |                   |

The Benchmark page tests Pico HTTP and DMX update performance.

Use it to compare:

- Single-channel updates
- Scene-sized batch updates
- Large batch updates
- Longer soak tests

The result panel shows:

- Throughput
- Effective DMX channel updates per second
- Average latency
- Median latency
- p95 and p99 latency
- Jitter
- Errors

Use **Export CSV** to save results for later comparison.

## 9. Backup and Import

Most setup data is stored as JSON files on the XAMPP server in the `data/` folder.

The UI also offers export/import buttons for:

- Fixture setup
- Groups
- Scenes
- Chaser setup
- Motion FX setup
- Fan Out setup
- GPIO setup

Use JSON export before large changes so a known-good setup can be restored later.

## 10. Clear Functions

There are two different clear actions:

| Action            | What it clears                             | When to use                                             |
|-------------------|--------------------------------------------|---------------------------------------------------------|
| DMX clear         | Live DMX output and the motion base buffer | Full reset of output and base position                  |
| Output-only clear | Live DMX output only                       | Black out output while keeping the stored motion center |

Use output-only clear when you want Motion FX to resume around the same stored center after the blackout.

## Troubleshooting

### The UI does not control the Pico

- Check that the Pico is powered and connected to WiFi.
- Open the Pico URL directly in a browser, for example `http://192.168.0.24/`.
- Make sure the base URL in the UI ends with `/`.
- Enable **Live send** on the Fixture Controller.

### Chaser capture does not contain the expected values

- Move or recall the values on the Fixture Controller first.

- Make sure the participating controls include the controls you want to capture.
- Save or reload the Fixture Controller setup if fixtures were changed recently.

## **Motion FX moves around the wrong center**

- Recall the desired scene first.
- On the Motion FX page, click the same scene in the scene toolbox.
- Then start the motion slot.

## **GPIO mapping does not react**

- Check `/gpio/status` or the status panel on the GPIO page.
- Confirm that the selected pin is not reserved or already used.
- Check pull mode and trigger edge.
- For ADC, use only GPIO26, GPIO27, or GPIO28.

## **A chaser has more than 32 steps**

The firmware supports 32 steps per chaser slot. Keep the UI preset within this limit before uploading to the Pico.